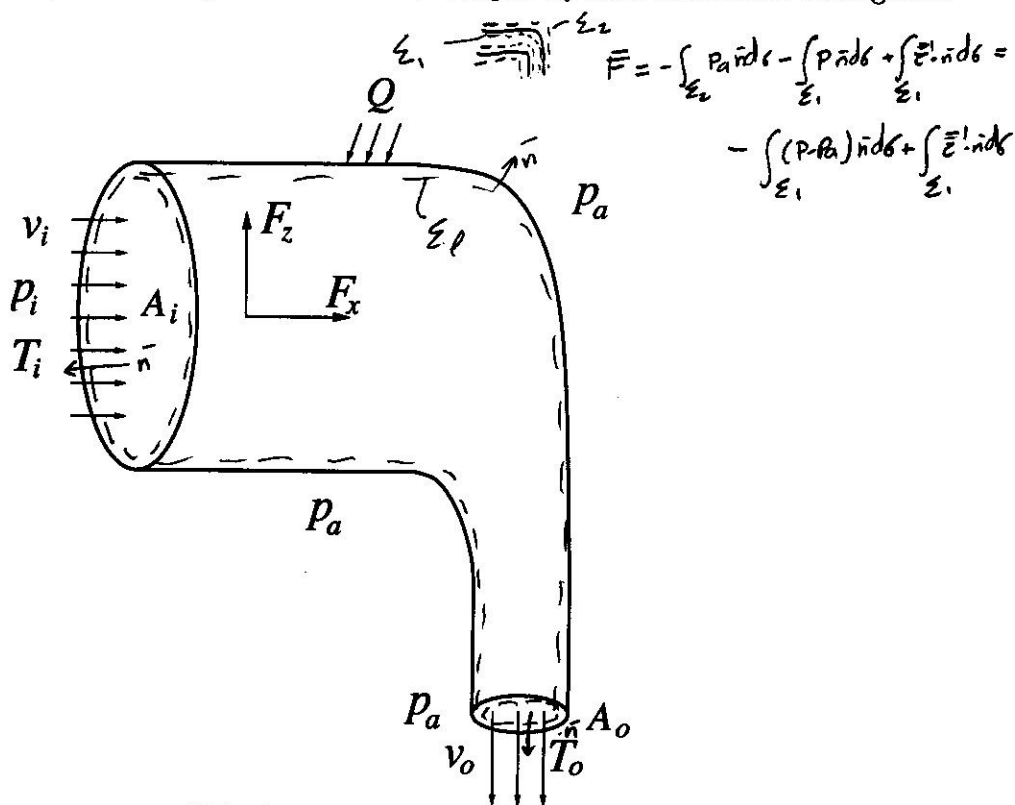


CL6

A liquid of density ρ and specific heat c flows through the elbow-shaped pipe shown below, discharging to the atmosphere as a vertical jet through an outlet section of surface area A_o , different from the value A_i at the inlet. Gravity is assumed to have a negligible effect on the fluid motion. Besides, uniform flow properties are assumed at the inlet and outlet sections, where readings from a thermometer give different values T_i and T_o for the temperature. The velocity at the outlet section is measured to be v_o and a manometer is used to determine the value p_i for the pressure at the inlet section. In terms of all of these quantities, determine:

1. The value of the velocity at the inlet v_i
2. The two components of the force (F_x, F_z) exerted on the pipe.
3. The amount of heat Q transferred per unit time to the liquid by heat conduction through the pipe walls.



$$\vec{F} = - \int_{\Sigma_2} p_a \vec{n} d\sigma - \int_{\Sigma_1} p \vec{n} d\sigma + \int_{\Sigma_1} \vec{\tau} \cdot \vec{n} d\sigma = - \int_{\Sigma_1} (p - p_a) \vec{n} d\sigma + \int_{\Sigma_1} \vec{\tau} \cdot \vec{n} d\sigma$$

CONT: $\rho v_i A_i = \rho v_o A_o \Rightarrow \boxed{v_i A_i = v_o A_o} \rightarrow v_i = v_o A_o / A_i$

MOMENTUM: $\frac{d}{dt} \int_{V_c} \rho \vec{v} dV + \int_{\Sigma_i + \Sigma_o + \Sigma_p} \rho \vec{v} \cdot \vec{n} d\sigma = - \int_{\Sigma_i + \Sigma_o + \Sigma_p} (p - p_a) \vec{n} d\sigma + \int_{\Sigma_i + \Sigma_o + \Sigma_p} \vec{\tau} \cdot \vec{n} d\sigma + \int_{V_c} \rho \vec{g} dV$

$$- \rho v_i A_i v_i \vec{e}_x = \rho v_o A_o v_o \vec{e}_y = (p_i - p_a) \vec{e}_x A_i - \vec{F}$$

$F_x = [\rho v_i^2 + (p_i - p_a)] A_i$
$F_y = \rho v_o^2 A_o$

ENERGY: $\frac{d}{dt} \int_{V_c} \rho \left(e + \frac{v^2}{2} \right) dV + \int_{\Sigma_i + \Sigma_o + \Sigma_p} \rho \left(e + \frac{v^2}{2} \right) \vec{v} \cdot \vec{n} d\sigma = - \int_{\Sigma_i + \Sigma_o + \Sigma_p} p \vec{v} \cdot \vec{n} d\sigma + \int_{\Sigma_i + \Sigma_o + \Sigma_p} \vec{v} \cdot \vec{\tau} \cdot \vec{n} d\sigma + \int_{V_c} \rho \vec{v} \cdot \vec{g} dV + \int_{\Sigma_i + \Sigma_o + \Sigma_p} k \nabla T \cdot \vec{n} d\sigma$

$\rho v_o A_o \left[\left(c T_o + \frac{v_o^2}{2} \right) - \left(c T_i + \frac{v_i^2}{2} \right) \right] \quad \downarrow \quad p_i v_i A_i - p_a v_o A_o$

Q

$$\boxed{Q = \rho v_o A_o \left[c (T_o - T_i) + \frac{v_o^2 - v_i^2}{2} \right] + (p_a - p_i) v_o A_o}$$